# Thoughts on the research plan

My answers/thoughts on the research plan. Text not AI generated.

## Understanding AI Pacing

Reading done:

- Pacing the Frontier: A Framework and Research Agenda. Fully read.
- Pacing the Frontier — Employee Statement. Read a few employee statements.

Answers to the questions:

*How does pacing differ from earlier proposals to pause or stop AI development?*

I don't have a lot of context for pausing/stopping proposals but I think stopping vs. pausing vs. pacing differ in how straightforward their end goal is and how realistic the proposition is given our real life context.

- Stopping AI development is the most straightforward in its aim and can be quite coarse in its implementation (I can think of clear, unambiguous metrics/strategies for stopping AI development: shut down existing labs and only retaining a specific set of their capacities + prevent new labs from entering the AI race). It's also what would most decisively prevent AI-related risks. However, it's probably the least realistic at this stage (would require coordinated action that neither governments nor markets have an incentive to pursue) + opportunity cost from lost AI benefits.
- Pausing AI development already requires more nuanced metrics (e.g. what is being paused?) and a more sophisticated implementation plan (questions like until when is the pause happening? Should we pause only some development or all development? Are the type of questions that become important). Just like with stopping though, this seems unrealistic given the position of the main actors (both governments and labs with leading capabilities will worry about other countries catching up)
- I think pacing is the most realistic strategy of the three as some of its aspects align well with the incentives of the labs (e.g. continuing development, having extra time to work on alignment and prevent AI takeovers, which is in their own interest). It's also the most nuanced of the three, has lots of tradeoffs (I like how carefully the pacing article goes through them; enjoyed in particular the false positives vs. false negatives vs. intrusiveness, pacing trilemma and overhang discussion) and requires operating under a lot of uncertainty.

*Consider the distinction between slowing capability development (slowing models getting better) and slowing deployment or diffusion (slowing how fast capability of deployed models improve, i.e. what everyday people experience)*

Ultimately model deployment is what we care about (how models are actually being used and what direct outcomes come from this) but in terms of actually affecting outcomes, slowing capability development

seems much more tractable than slowing deployment (clearer benchmarks for “capabilities” than “diffusion” + earlier in the chain, i.e. if you target capability pacing you will likely prevent a wide range of the deployment risks that come from these capabilities; if you’re just targeting deployment itself, you’re always playing catch up with the capabilities).

## Frontier Lab Perspectives

Reading done:

- We Must Pace the Frontier. Fully read.
- Priorities and Principles for Effective Third-Party Assessments. Fully read.

Answers to the questions:

*What do labs appear willing to accept voluntarily?*

Anthropic (3 main points in the article):

1. “Embedded evaluators” proposal (measure already adopted by banking industry): willing to work with permanent independent evaluators (provide them with employee-like status and allow them to publish their own findings). The article calls for other labs to adopt this practice as well but it seems like Anthropic is willing to adopt this unilaterally, maybe regardless of broader adoption.
2. Participation in coordination among pacing within democratic countries: happy with complying to industry-level binding safety regulations + industry-level information sharing (provided the correct legal framework is in place) + willing to limit training compute + willing to implement restrictions on harmful use of AI models.
3. Participation/willingness to comply with international coordination regulation: restrictions on harmful use, global-level standards for AI safety testing, agree to be subject to speed limits for recursive self improvement

Open AI:

Third-party evaluators: fairly similar in goals to Anthropic’s proposal (safety focused evaluations, evaluators publish findings, agree with industry-level coordination), however proposes independent *external* assessors (not employee-like level) with agreed upon, more limited scope than Anthropic.

*Where do their incentives might diverge from effective oversight?*

Areas where, in principle, the incentives from a frontier lab would not naturally align with oversight:

- Publication of unfavorable findings from independent evaluators
- Non-employees with continuous access to the company
- Labs subjecting themselves to investigations after incidents
- The capability checkpoint idea from Amodei (“if models have capability X, then they need to be accompanied by certifications of alignment properties Y and Z”)

*Do different frontier labs have materially different positions?*

Not clear from the articles only (Amodei's article is a coherent vision for pacing while Open AI's article was specifically for third party evaluations, so they don't cover the same topics). Would be interested to know more.

## Verification, Monitoring & Enforcement Mechanisms

Reading done:

- Verification Methods for International AI Agreements. Read executive summary.
- Verifying International Agreements on AI: Six Layers of Verification. Created and read a summary with AI.
- Flexible Hardware-Enabled Guarantees for AI Compute. Read executive summary.

Answers to the questions:

| | What it can actually verify or enforce | Which parties need to cooperate | How readily it could be circumvented | Whether it meaningfully slows capability development rather than merely satisfying a measurable restriction |
|---|---|---|---|---|
| Independent third-party audits | Training practices, investigate incidents, compliance with agreed rules. But can't "prevent" violations from happening in the first place | AI Labs and independent auditors (potentially regulators?) | Moderately, depends on AI lab's willingness to cooperate (e.g. employee-like access makes it less likely to circumvent) | Not sure |
| Cloud/ infrastructure monitoring | Compute usage and workload. Can enforce limits | Cloud providers, datacentre operators | Moderately, for e.g. via alternative providers | Yes, can effectively constrain scaling |
| Hardware-based controls | Verify chip usage, enforce usage limits | Chip manufacturers (potentially unilaterally?) | Not easy to circumvent, would require tampering | Can constrain scaling based on physical compute but not algorithmic efficiency |
| Cryptographic verification | Verify compliance with specified restrictions without revealing sensitive info | AI Labs, infrastructure operators | Moderately, e.g. perform forbidden activities elsewhere | Unlikely? |
| Research workflow/model lifecycle controls | Slowing the pace of AI research | AI Labs, regulators, independent auditors | Moderately, depends on willingness of labs + how strictly is regulation enforced | Quite likely to prevent capability development |

# Thoughts on the direction of the project

1. If we're not going to focus on some particular verification procedure but instead classes of verification procedures, it would be helpful to have a much clearer taxonomy of the verification procedures out there, what they have in common (what forms a "class"). From there we can start thinking about risks and incentives to deviate.
2. Once we have the taxonomy – how do we pick which classes of procedures to focus on? Do we want to focus on the procedures we think are the most likely to lead to effective pacing, or do we

want to focus on the procedures we think are the most invasive (or least invasive), or do we want to focus on the "riskiest" (most likely to distort behaviour in negative ways)? Or do we want to be comprehensive?

3. How much do we want to take into account empirical features of existing labs (e.g. the "personality" of Anthropic/OpenAI/Google DeepMind) vs. thinking about the theoretical incentives of labs? Same question for other actors (tindependent auditors, governments etc.)

Questions:

- Things I highlighted
- Clarify overlap/difference with question 4: "How do the incentives of bound parties change across the lifecycle? To what extent can different actors reliably predict the behavior of other actors throughout the lifetime of a pacing intervention? How load-bearing are these predictions of behavior going to be for coordination of pacing interventions?"